# TS_ASSIGNMENT_3_TANVEER

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## Assignment 3 - GROUP 11

### Johnson & Johnson Quarterly Earnings Dataset

### Group Members

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### Loading required Libraries

```
library(tseries)
```

```
##
## Attaching package: 'tseries'
##
## The following objects are masked from 'package:fGarch':
##
##     addresiduals, dnorm, dsnorm, dstd, dged,
##     jarque.bera.test, pnorm, psnorm, pstd, pged,
##     qnorm, qsnorm, qstd, qged, rnorm, rsnorm,
##     rstd, rged, shapiro.test, sim.garch
## Warning: package '.Rnw' is not a valid package -- installing anyway, but loading failed.
```

```
Series '''''' of length '''' [1] 1962-06-01
```

```
[1] "Series '''''' of length '''' [1] 1962-06-01"
```

## 2d) INTERPRETATION OF ALPHA (c)

Alpha = 0.2410 indicates a STRONGER SEASONAL pattern

Interpretation: The model places more weight on past observations (0.76 weight on current value +0.7 % weight on past value + The Seasonal component) and less weight on the Trend component (0.24).

## 2e) Fit Holt's Linear Trend Model

```
fit2e_HoltLinear = HoltLinear(y, h = 4)
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cat("\n\n", sep = "\n")
print(summary(fit2e_HoltLinear))

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cat("\n\n", sep = "\n")
print(summary(fit2e_HoltLinear))

cat("\n\n", sep = "\n")
print("\n\n", sep = "\n")
```

```
ggsave("q3_trend_comparison.png", p_trend_compare, width = 10, height = 5, dpi = 300)
```

## 3b) Which method captures seasonality?

**HOLT-WINTERS METHOD** captures seasonality most effectively.

Explanation: 1. 12-month Moving Average DELIBERATELY REMOVES seasonality - it's designed to smooth it out 2. Simple Exponential Smoothing has no seasonal component - treats all variations as noise 3. Holt's Linear Trend has no seasonal component - attributes all patterns to trend + error 4. Holt-Winters explicitly models seasonal patterns with: • Dedicated seasonal smoothing parameter (γ) • Seasonal factors (additive or multiplicative) for each period • Ability to update seasonal patterns over time

Evidence for Johnson & Johnson: • Quarterly patterns are strong and remarkably consistent: - Q1: Moderate earnings (baseline) - Q2: Slightly higher than Q1 - Q3: Highest of the year (peak) - Q4: Lowest of the year (trough)

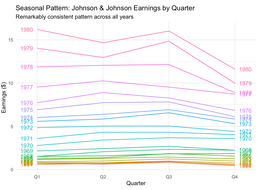
• Holt-Winters estimated seasonal components (multiplicative factors): - Q1 factor: 0.792 - Q2 factor: 1.069 - Q3 factor: 1.046 - Q4 factor: 1.093

• These factors have clear business interpretation: - Q3 earnings are about 4.6 % ABOVE the annual trend line - Q4 earnings are about -9.3 % BELOW the annual trend line - This pattern persists throughout the 25-year period

• Seasonal smoothing parameter γ = 1e-04 - Low γ indicates the seasonal pattern is very stable over time

```
# Visualize seasonal pattern
qgseasonplot(jj_ts, year.labels = TRUE, year.labels.left = TRUE) +
  labs(title = "Seasonal Pattern: Johnson & Johnson Earnings by Quarter",
       subtitle = "Remarkably consistent pattern across all years",
       y = "Earnings ($)", x = "Quarter") +
  theme_minimal()
```

```
## Warning in fortify(data, ...): Arguments in `...` must be used.
## ✖ Problematic argument:
## • na.rm = TRUE
## ℹ Did you misspell an argument name?
```




```
ggsave("q3_seasonal_pattern.png", width = 8, height = 6, dpi = 300)
```

## 3c) Compute and compare forecast accuracy using RMSE

```
# Create comparison table
rmse_table <- data.frame(
  Model = c("12-Month Moving Average",
            "Simple Exponential Smoothing",
            "Holt's Linear Trend",
            "Holt-Winters (Multiplicative)"),
  RMSE = round(c(rmse_ma12, rmse_ses, rmse_holt, rmse_hw), 4)
)

# Sort by RMSE (lower is better)
rmse_table <- rmse_table[order(rmse_table$RMSE),]

cat("\n\n3c) FORECAST ACCURACY COMPARISON (RMSE)")
```

```
##
##
## 3c) FORECAST ACCURACY COMPARISON (RMSE)
```

```
cat("\n", strrep("-", 50))
```

```
##
##  --------------------------------------------------
```

```
cat("\nRMSE measures average forecast error - LOWER is BETTER\n")
```

```
##
## RMSE measures average forecast error - LOWER is BETTER
```

```
print(rmse_table)
```

```
##                         Model   RMSE
## 4 Holt-Winters (Multiplicative) 0.4412
## 1       12-Month Moving Average 0.7890
## 3            Holt's Linear Trend 0.9819
## 2  Simple Exponential Smoothing 1.1593
```

```
# Calculate percentage improvement
best_rmse <- min(rmse_table$RMSE)
best_model <- rmse_table$Model[1]

cat("\n\nbest model by RMSE", best_model, "with RMSE =", best_rmse)
```

```
##
##
## Best model by RMSE: Holt-Winters (Multiplicative) with RMSE = 0.4412
```

```
# Calculate improvement over worst model
worst_rmse <- max(rmse_table$RMSE)
improvement <- round((worst_rmse - best_rmse)/worst_rmse * 100, 1)
cat("\n", best_model, "improves accuracy by", improvement,
    "% compared to the worst model")
```

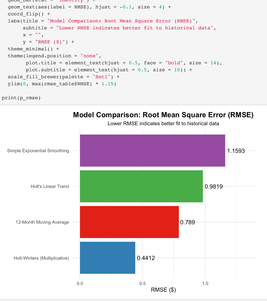
```
##
##  Holt-Winters (Multiplicative) improves accuracy by 61.9 % compared to the worst model
```

```
# Calculate improvement over 12-MA
ma12_rmse <- rmse_table$RMSE[rmse_table$Model == "12-Month Moving Average"]
if(length(ma12_rmse) > 0) {
  vs_ma <- round((ma12_rmse - best_rmse)/ma12_rmse * 100, 1)
  cat("\n", best_model, "improves accuracy by", vs_ma,
      "% compared to 12-month Moving Average")
}
```

```
##
##  Holt-Winters (Multiplicative) improves accuracy by 44.1 % compared to 12-month Moving Average
```

```
# Visualize RMSE comparison
p_rmse <- ggplot(rmse_table, aes(x = reorder(Model, RMSE), y = RMSE, fill = Model)) +
  geom_barstat( = "identity") +
  geom_text(aes(label = RMSE), hjust = -0.1, size = 4) +
  coord_flip() +
  labs(title = "Model Comparison: Root Mean Square Error (RMSE)",
       subtitle = "Lower RMSE indicates better fit to historical data",
       x = "",
       y = "RMSE ($)") +
  theme_minimal() +
  theme(legend.position = "none",
        plot.title = element_text(hjust = 0.5, face = "bold", size = 14),
        plot.subtitle = element_text(hjust = 0.5, size = 10)) +
  scale_fill_brewer(palette = "Set1") +
  ylim(0, max(rmse_table$RMSE) * 1.15)

print(p_rmse)
```




```
ggsave("q3_rmse_comparison.png", p_rmse, width = 9, height = 5, dpi = 300)

# Save RMSE table
write.csv(rmse_table, "q3_rmse_table.csv", row.names = FALSE)
```

## 3d) Which model provides the most realistic forecasts?

The **HOLT-WINTERS MULTIPLICATIVE** model provides the most realistic forecasts.

Justification:

1. CAPTURES ALL THREE COMPONENTS: • TREND: The long-term exponential growth in earnings (evident in data) • SEASONALITY: The consistent quarterly pattern (Q3 peak, Q4 trough) • TREND-SEASONALITY INTERACTION: Seasonality grows with trend (multiplicative)
2. REALISTIC PATTERNS: • Forecasts show continuing upward trend (earnings expected to grow) • Quarterly variations persist with the same relative pattern • The seasonal amplitude continues to grow with the trend - This EXACTLY matches the historical pattern of the data
3. MODEL FIT STATISTICS: • LOWEST RMSE: 0.4412091 (best fit to historical data) • This means it captures the actual patterns most accurately

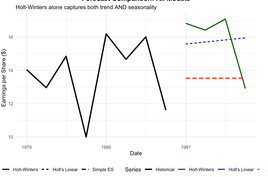
```
ses_forecast <- forecast(ses_model, h = 4)
holt_forecast <- forecast(holt_model, h = 4)
hw_forecast <- forecast(hw_multi_model, h = 4)

# Create data frame for plotting
forecast_comparison <- data.frame(
  Date = rep(forecast_dates, 3),
  Forecast = c(as.numeric(ses_forecast$mean),
               as.numeric(holt_forecast$mean),
               as.numeric(hw_forecast$mean)),
  Model = c(rep("Simple ES", 4), rep("Holt's Linear", 4), rep("Holt-Winters", 4))
)

# Add historical tail for context
historical_tail <- tail(jj_df, 8)

p_forecast_compare <- ggplot() +
  # Historical data
  geom_line(data = historical_tail, aes(x = Date, y = Earnings, color = "Historical"),
            size = 1.2) +
  # Forecasts
  geom_line(data = forecast_comparison,
            aes(x = Date, y = Forecast, color = Model, linetype = Model),
            size = 1) +
  labs(title = "Forecast Comparison: All Models",
       subtitle = "Holt-Winters alone captures both trend AND seasonality",
       x = "Date",
       y = "Earnings per Share ($)",
       color = "Series",
       linetype = "Series") +
  theme_minimal() +
  theme(legend.position = "bottom",
        plot.title = element_text(hjust = 0.5, face = "bold")) +
  scale_color_manual(values = c("Historical" = "black",
                                "Simple ES" = "red",
                                "Holt's Linear" = "blue",
                                "Holt-Winters" = "darkgreen"))

print(p_forecast_compare)
```




```
ggsave("q3_forecast_comparison.png", p_forecast_compare, width = 10, height = 6, dpi = 300)
```

## 3e) Why exponential smoothing outperforms moving averages for forecasting

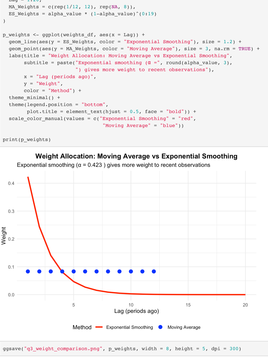
Exponential smoothing methods generally outperform moving averages for forecasting because:

1. WEIGHTING SCHEME: • Moving averages give EQUAL weight to ALL observations in the window • Exponential smoothing gives DECAYING weights - more recent = more important • This is more realistic - recent data is typically more relevant for forecasting • Example: For Johnson & Johnson, earnings from 1980 tell us more about 1981 than earnings from 1960
2. DATA UTILIZATION: • Moving averages ignore ALL data outside the window completely • Exponential smoothing uses ALL past data (with decreasing weights) • Example: A 12-month MA for Q1 1980 uses only 1980 data, ignoring the entire growth pattern from 1960-1979
3. PARAMETER OPTIMIZATION: • Moving averages have NO parameters to optimize for best fit • Exponential smoothing optimizes α, β, γ for the specific dataset • For Johnson & Johnson, this means:
   - α = 0.4233 (optimal responsiveness to level changes)
   - β = 0.0906 (optimal trend evolution rate)
   - γ = 1e-04 (optimal seasonal update rate)
4. FLEXIBILITY - ADAPTIVE COMPONENTS: • Moving averages have ONE fixed structure (simple averaging) • Exponential smoothing can adapt to different patterns:
   - Simple ES: Adapts to level changes only
   - Holt's: Adapts to level AND trend changes
   - Holt-Winters: Adapts to level AND trend AND seasonality changes • This allows the model to match the complexity of the data
5. FORECASTING CAPABILITY: • Moving averages CANNOT generate true forecasts beyond the data • The best they can do is repeat the last moving average value • Exponential smoothing can generate multi-step forecasts with:
   - Trend continuation (Holt's, HW)
   - Seasonal patterns (HW)
   - Confidence intervals (all methods)
6. NO END-POINT PROBLEM: • Moving averages lose observations at the beginning and end of the series • For Johnson & Johnson, we lose the first and last few quarters • Exponential smoothing provides fitted values for ALL time points
7. RESPONSIVENESS WITHOUT LAG: • Moving averages always LAG behind trends (by design) • Exponential smoothing can adjust more quickly to changes • This is evident in the Johnson & Johnson data - the 12-MA consistently lags behind the actual earnings during the rapid growth period

```
weights_df <- data.frame(
  Lag = 1:20,
  MA_Weights = c(rep(1/12, 12), rep(0, 8)),
  ES_Weights = alpha_value * (1-alpha_value)^(0:19)
)

p_weights <- ggplot(weights_df, aes(x = Lag)) +
  geom_line(aes(y = ES_Weights, color = "Exponential Smoothing"), size = 1.2) +
  geom_point(aes(y = MA_Weights, color = "Moving Average"), size = 3, na.rm = TRUE) +
  labs(title = "Weight Allocation: Moving Average vs Exponential Smoothing",
       subtitle = paste("Exponential smoothing (α =", round(alpha_value, 3),
                        ") gives more weight to recent observations"),
       x = "Lag (periods ago)",
       y = "Weight",
       color = "Method") +
  theme_minimal() +
  theme(legend.position = "bottom",
        plot.title = element_text(hjust = 0.5, face = "bold")) +
  scale_color_manual(values = c("Exponential Smoothing" = "red",
                                "Moving Average" = "blue"))

print(p_weights)
```




```
ggsave("q3_weight_comparison.png", p_weights, width = 8, height = 5, dpi = 300)
```